Md Abdullah Hossien Molla

Bachelor of Computer Science
Independent University, Bangladesh
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Profile

Final-year Computer Science student with hands-on research expertise in Computer Vision, Deep Learning, and Applied Machine Learning. Author of 4 peer-reviewed publications (including 2 as First Author) spanning object detection, clinical ML, and assistive robotics. Experienced across the full research lifecycle from dataset curation and PyTorch/YOLO benchmarking to peer-reviewed manuscript publication. Seeking a research-oriented Master's program to contribute to laboratory research in visual computing and deployable AI systems.

Education

- **Bachelor of Science in Computer Science** — *2022–2026*
  Independent University, Bangladesh — CGPA: 3.64 / 4.00
- **Higher Secondary Certificate (HSC)** — *2020*
  Govt. Science College, Tejgaon — GPA: 5.00

Publications

- **Understanding Challenges of Liver Data Governance for Mobile App-Based Liver Care in Bangladesh**
  [Co-Author] | HCII 2026, Springer LNCS, pp. 120–138 | DOI: 10.1007/978-3-032-30549-7_7

  Investigated data governance gaps in mobile liver-care apps in Bangladesh. Conducted stakeholder interviews to identify privacy and data-sharing challenges. Proposed governance recommendations for low-resource healthcare settings.

- **Safety-Enhanced Machine Learning for Fetal Health Classification Using Cardiotocography Data**
  [First Author] | ICMLAS 2026, IEEE | DOI: 10.1109/ICMLAS67792.2026.11483766

  Developed safety-aware ML models for fetal health classification using CTG data. Applied feature engineering and evaluation metrics to reduce misclassification risk. Benchmarked multiple classifiers to identify the most reliable clinical approach.

- **Braille for All: A $70 AI-Powered Braille Tutoring Robot That Erases the Literacy Gap with a Touch and a Voice**
  [Co-Author] | CCE 2026, IEEE, Mexico City | In press

  Co-designed a low-cost ($70) AI-powered Braille tutoring robot for visually impaired learners. Integrated voice and touch feedback for self-paced learning in resource-limited settings. Contributed to system design and evaluation of the assistive learning workflow.

- **Road Condition Detection on Bangladeshi Roads: A Benchmark of YOLO Architectures**
  [First Author] | ICSIE 2026 | Accepted, in press

  Trained and benchmarked YOLOv5s, v8s, v10s, and v11s on the Bangladesh Road Dataset, comprising 11,750 annotated instances across six road conditions derived from 4,254 augmented training images. Achieved 71.1% mAP50 with the best-performing architecture for road-condition detection. Proposed a benchmark for infrastructure monitoring in developing-country road conditions.

Research Experience

- **Research Assistant** — *Aug 2025 – May 2026*
  On-Campus — Independent University, Bangladesh
  - Conducted literature reviews to identify research gaps and inform study design.
  - Designed and executed field data collection, including conducting interviews with participants to gather primary research data.
  - Contributed to research writing and manuscript preparation for peer-reviewed publications.
  - Worked alongside faculty and co-authors on multiple research projects spanning applied ML and HCI domains.

- **Teaching Assistant — Dept. of CSE** *Nov 2024 – Jan 2026*
  On-Campus Independent University, Bangladesh
  - Supported faculty with course delivery, grading, and student evaluation for 100+ students per term.
  - Ran office hours and review sessions to help students work through foundational CS concepts.

Selected Research Projects

- **Road Condition Detection on Bangladeshi Roads — Computer Vision System**
  - Trained and benchmarked four YOLO variants (v5s–v11s) on 11,750 annotated instances across 6 road conditions using PyTorch, reaching 71.1% mAP50.
  - Set up the full training pipeline with reproducible dataset splits and evaluation metrics.
  - *Tech: Python, PyTorch, Ultralytics YOLO, OpenCV, Google Colab*

- **GloHealth AI — ML-Backed Health Symptom Reporting System**
  - Engineered symptom features and trained Scikit-learn and XGBoost classifiers for disease prediction.
  - Integrated explainable ML output behind a Flask REST API, so end users see why a condition was predicted.
  - *Tech: Python (Flask), Scikit-learn, XGBoost, Supabase, PostgreSQL*

Technical Skills

Programming: Python, C/C++, JavaScript, SQL.
Machine Learning: PyTorch, Ultralytics YOLO, Scikit-learn, XGBoost, OpenCV.
Data Analysis: Pandas, NumPy, Matplotlib, Seaborn.
Tools & Platforms: Git/GitHub, LaTeX, Jupyter, Google Colab, Docker, PostgreSQL.
Research: Literature review, experimental design, field data collection, interview-based research, academic writing.

Honors & Achievements

- Merit-based Undergraduate Scholarship, Independent University, Bangladesh (awarded for academic standing).
- Dean's Merit List recognition for consistent academic performance.
- International FIDE Rapid Chess Rating.
- Event Coordinator, July Memorial FIDE Chess Championship 2025 (IUB) — coordinated FIDE officials, venue logistics, and 100+ participant registrations.

References

- **Mohammad Shidujaman, PhD**
  Assistant Professor, Dept. of CSE, School of Engineering, Technology and Sciences, Independent University, Bangladesh
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- **Md Tarek Habib, PhD**
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